

with materials that not only provide better protection but store energy and monitor the health status of our soldiers. Science fiction writers can and do have a lot of fun imagining such things.

Imagine bullets that pack more computer power than the largest supercomputer in existence today, allowing them to perform real time image analysis of their surroundings and communicate with weapons tracking systems to acquire and navigate to targets with greater precision and control. Or imagine UAVs equipped with chemical sensors based on nanotechnology capable of pinpointing a single molecule out of billions and producing incredible photo resolutions

In fact nanotechnology even offers vivid possibilities for creative mass murder. For example, for some reason one of the most frequent flights of fancy is the programmable genocide germ that replicates freely and kills people who have certain DNA patterns. Such a weapon is possible, but it is dangerous to its creators as well. How does one go about overcoming and attacking a nanotechnic foe that doesn't depend on people to run his war machine.



Inhumanly quick

Killing is not difficult; what is difficult is to kill with impunity when one's enemy is well armed and numerous. War is more often a contest to suppress one enemy's capabilities before the enemy can suppress ones. This doesn't leave much room for fancy swordplay or gothic revenge scenarios in serious combat. An actual nanotechnic war, if one ever occurs, is likely to be inhu-

manly fast and enormously destructive.

The possibility of the development of assembler-based molecular nanotechnology presages a potential for disruption and chaos in the world system and is a cause of concern. What is listed below might turn out to be the events which might lead to a third and the last World War.

1. The prospect of revolutionary advances in military capabilities will stimulate competition to develop and apply the new technologies toward war preparations. No country will be able to afford to fall behind, as that would imply an intolerable security risk.
2. A race to develop early military applications of molecular manufacturing might yield sudden breakthroughs, leading to the emergence of new and unfamiliar threats, and provoking political and military reactions which further reinforce a cycle of competition and confrontation.